

Smart Helmet with Alcohol Detection

MINIOR PROJECT REPORT

Submitted in Partial Fulfilment of the Requirements for the Degree of

Electronics and Communication Engineering By

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CERTIFICATE

This is to certify that the minor project entitled “**Smart Helmet with Alcohol Detection**” that is being submitted by **K. Vinay Shankar, V. Varun sai, M. Lalith Krishna** bearing Regd.No.191FA05037,191FA05062, 201LA05001 in partial fulfilment for the award of II year II semester B. Tech degree in Electronics and Communication Engineering to Vignan’s Foundation for Science Technology and Research, is a record of work carried out by them under the guidance of DR.PACHIYANAN, MR.JANAKI RAM ,DR.SHARAD KUMAR TIWARI , MRS.SHIVA DURGA, of ECE Department.

Signature of the faculty guide

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ABSTRACT

An accident is an unexpected, unusual, unintended external action which occurs in particular time and place. Carelessness of the driver is the major factor for accident. The government has made rules that rider should compulsory wear the helmet and not consume alcohol and drive. Still the riders do not obey the rules. These accidents are caused due to negligence of the rider. Not wearing the helmet causes the rider with head injuries which may lead to death of the rider. In order to overcome this an intelligent system, smart helmet is proposed, it detects the helmet and also the alcohol present in rider's breath. This system has a pair of transmitter and receiver, the transmitter is placed in the helmet and the receiver is placed at the bike ignition. There are different sensors to ensure the helmet is on the head. These vibration sensors are placed in helmet where the probability of hitting is more. An alcohol sensor is placed near mouth of the rider.

The alcohol sensor detects the presence of alcohol in rider's breath. The data of the detection of helmet and alcohol is coded with RF encoder and then transmitted through radio frequency transmitter. The receiver at the bike receives the data and the data is decoded using RF decoder. The result of presence of helmet and the alcohol detection is analyzed on the smart phone. The proposed system will be so designed that if one of the two conditions are violated then also the bike won't start. The bike will start only if the both conditions are followed. This smart helmet will help the rider to compulsory wear helmet and restrict drink and drive condition. MCU controls the function of relay and the ignition, it control the engine through a relay and a relay interfacing circuit.

INTRODUCTION

A traffic accident is defined as any vehicle accident occurring on a public highway (i.e. originating on, terminating on, or involving a vehicle partially on the highway). These accidents therefore include collisions between vehicles and animals, vehicles and pedestrians, or vehicles and fixed obstacles. In higher-income countries, road traffic [1] accidents are already among the top ten leading causes of disease burden in 1998 as measured in DALYs (disability-adjusted life years). In less developed countries, road traffic accidents were the most significant cause of injuries, ranking eleventh among the most important causes of lost years of healthy life. In Indian road system, widening of the road is not an alternative solution to avoid traffic in such a cities [2]. The problems with state drunk driving control systems can be solved in many ways. The most effective will follow several principles: They will invest authority and responsibility in people and organizations at all levels, local to national, because drunken driving control [6] requires action at all levels. They will operate in the public eye, using the media to report on problems and solutions, because ultimate decisions on priorities and resources to control drunk driving must have public support. They will not promise instant solutions based on a single action but rather will take steady steps towards long-term improvement. And they will establish mechanisms for identifying and solving problems rather than attempting to apply one-size-fits-all methods.

Circuit Diagram:

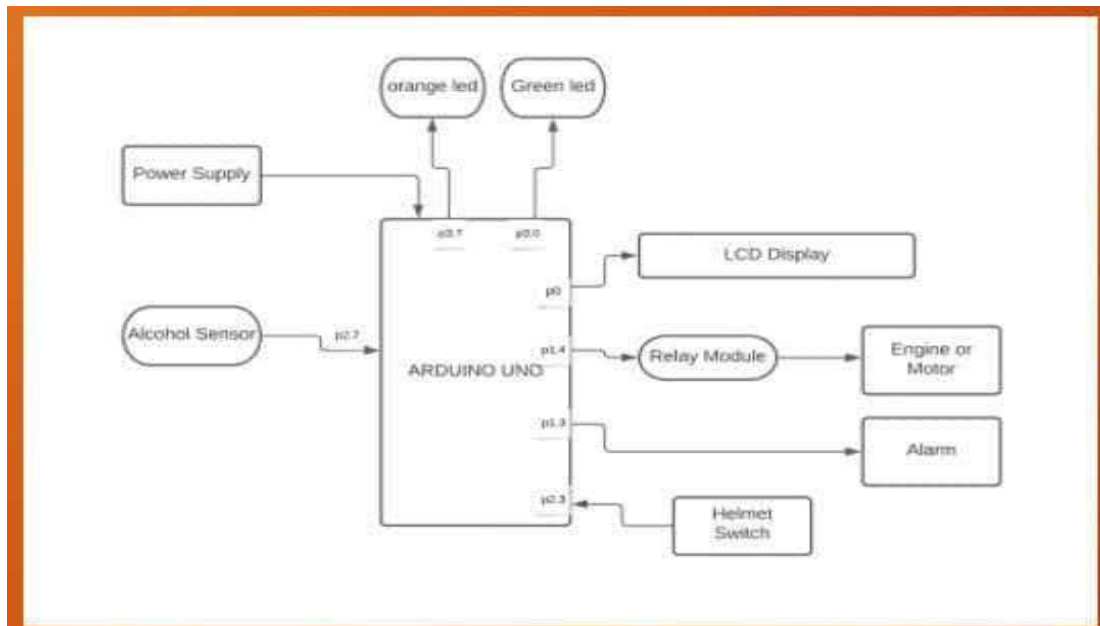


FIG-1

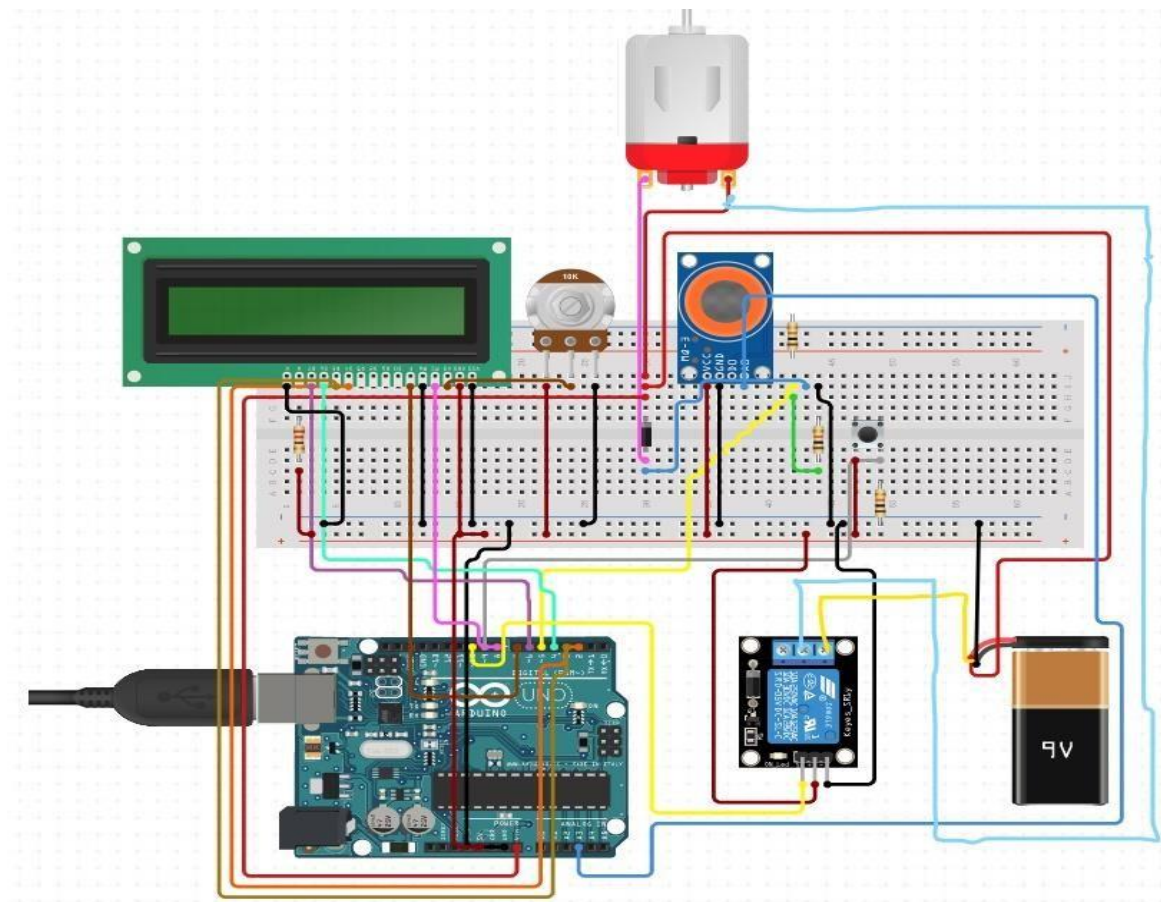


FIG-2

HARDWARE REQUIREMENTS :

1. Arduino UNO
2. Alcohol Sensor (MQ-3)
3. Relay Module
4. Push Button Switch
5. DC Motor
6. LCD (16x2 display)
7. Connectors/Jumpers
8. 9volt Battery
9. Buzzer & Bulb

Arduino Uno:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino senses the environment by receiving inputs from many sensors, and affects its surroundings by controlling lights, motors, and other actuators. You can tell your Arduino what to do by writing code in the Arduino programming language and using the Arduino development environment.

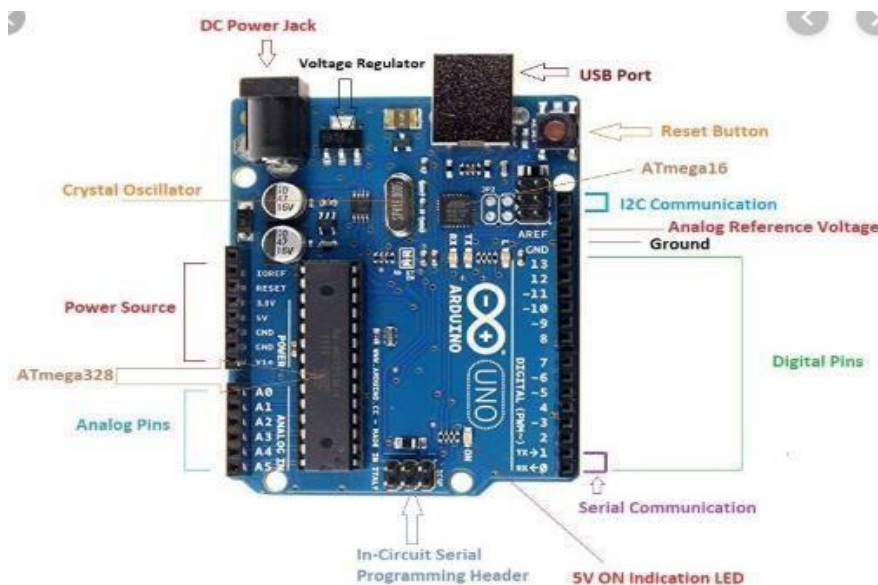


FIG-3

Alcohol sensor(MQ3):

A breadboard is a solderless device for temporary prototype with electronics and test circuit designs. Most electronic components in electronic circuits can be interconnected by inserting their leads or terminals into the holes and then making connections through wires where appropriate. The breadboard has strips of metal underneath the board and connect the holes on the top of the board. The metal strips are laid out as shown below. Note that the top and bottom rows of holes are connected horizontally and split in the middle while the remaining holes are connected vertically .

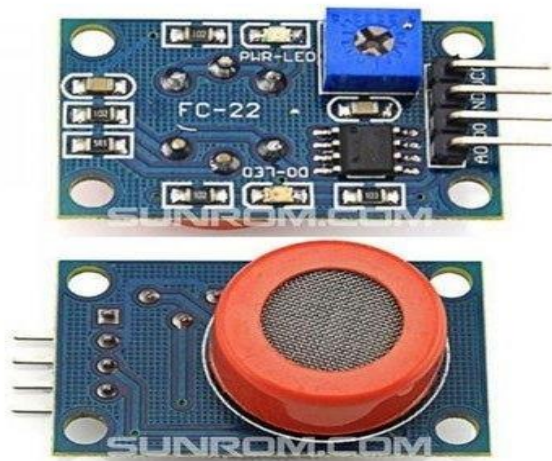


FIG-4

Relay Module:

The relay module is an electrically operated switch that allows you to turn on or off a circuit using voltage and/or current much higher than a microcontroller could handle. There is no connection between the low voltage circuit operated by the microcontroller and the high power circuit. The relay protects each circuit from each other. The each channel in the module has three connections named NC, COM, and NO. Depending on the input signal trigger mode, the jumper cap can be placed at high level effective mode which 'closes' the normally open (NO) switch at high level input and at low level effective mode which operates the same but at low level input.



FIG-5

DC Motor:

A **DC motor** is any of a class of rotary electrical motors that converts direct current electrical energy into mechanical energy. The most common types rely on the forces produced by magnetic fields. Here we used this as a engine motor.



FIG-6

BLOCK DIAGRAM:

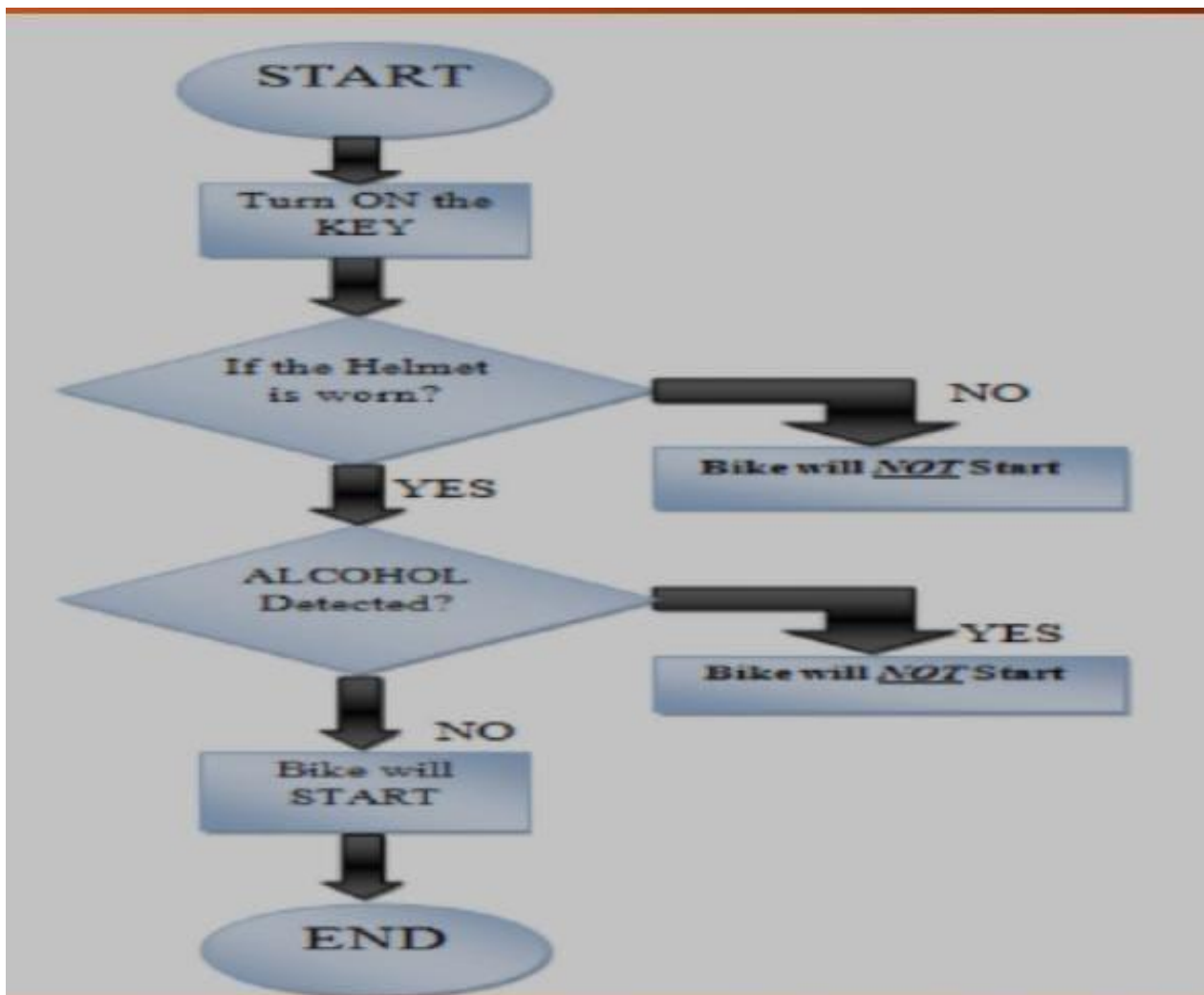


FIG-7

At first turn on the key then it detect if the person helmet is wear or not wear, after if helmet is weared then in the helmet there is a alcohol sensor detects if the person is alcoholic or not if the person is alcoholic then bike will not start if not detect then bike will start.

CIRCUIT CONNECTIONS:

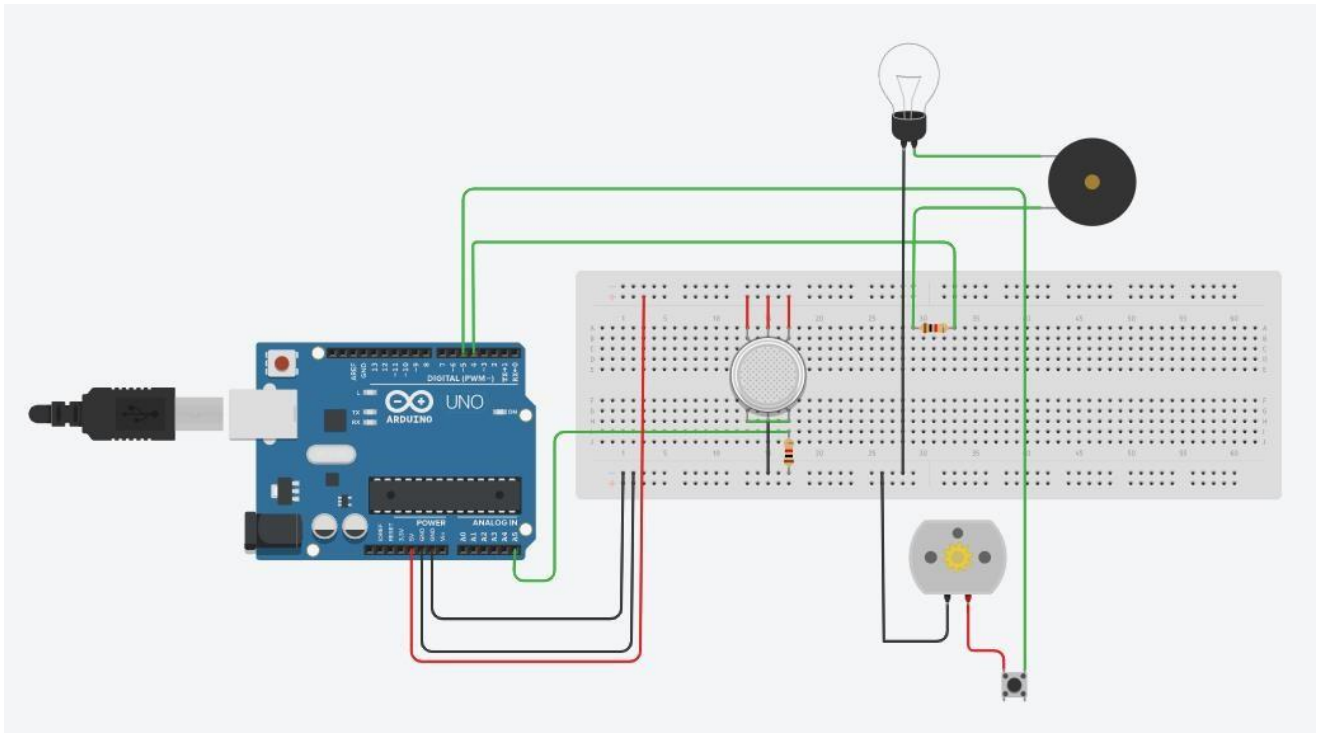


FIG-8

ARDUINO CODE :

```
#include <Liquid Crystal. h>
```

```
Liquid Crystal lcd (8, 9, 10, 11, 12, 13); //RS,EN,D4,D5,D6,D7
```

```
#define gas 5
```

```
#define engine 6
```

```
#define gnd 3
```

```
#define relay 7 int
```

```
alc=0, button=0;
```

```
void setup ()
```

```
{
```

```

Serial. Begin (9600);    // initialize serial communications at 9600 bps:

Pin Mode (gas, INPUT);

Pin Mode (engine, INPUT_PULLUP);

Pin Mode (relay, OUTPUT);

Pin Mode (gnd, OUTPUT);

Digital Write (relay, HIGH);

Digital Write(gnd, LOW);  lcd.

Begin (16, 2) ;//initializing LCD

lcd. Set Cursor (0,0);  lcd.

print("Smart Vehicle");

  lcd. Set Cursor (0,1);lcd.

print("Protection"); delay (3000);

lcd. clear (); lcd. Set Cursor (0,0);

lcd. print ("press key....");

delay(1000);

} void loop() {

if(alc==HIGH)

{

  if(button==LOW)

  { lcd. clear();

    Digital Write(relay, LOW);

    lcd. set Cursor(0,1);    lcd.

```

print("safe drive...");

delay(2000);

 }

}

else

{

 Digital Write(relay, HIGH);

 lcd. set Cursor(0,0);

 lcd. print("ALCOHOL DETECTED");

 delay(1000);

 lcd. clear();

 }

}

Experimentation Results :

The designed Smart helmet ensures the safety of the rider by making it necessary to wear helmet, and also ensures that the rider hasn't consumed alcohol more than the permissible limit. If any of these primary safety rules are violated, the proposed system will prevent the biker from starting the bike. Hence sensor detects the alcohol then engine stops even though helmet was wearred.

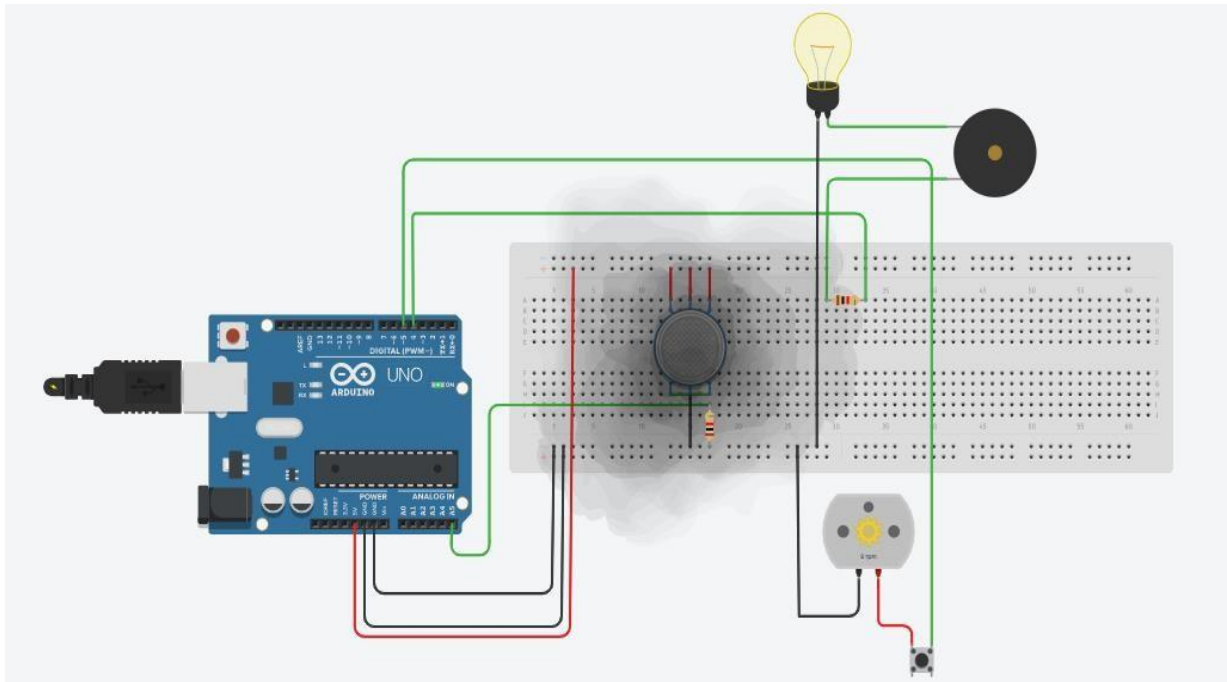


FIG-9

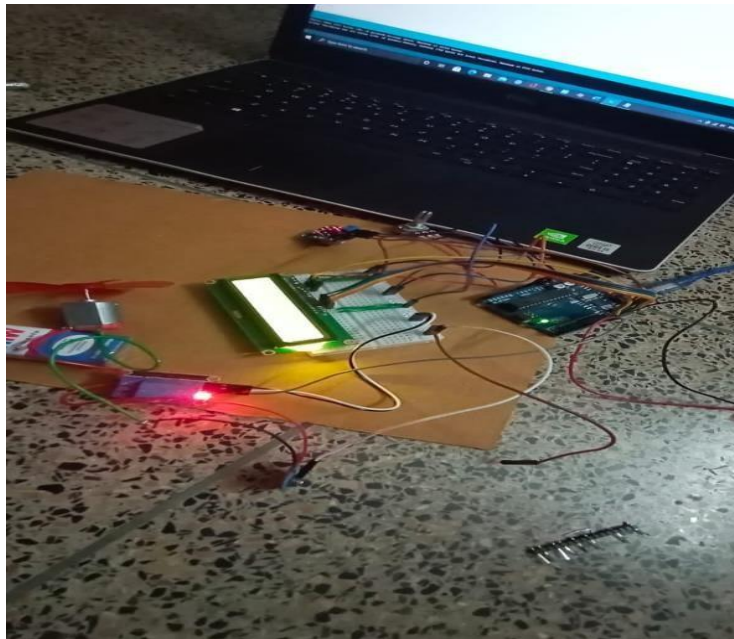


FIG-10

FUTURE DEVELOPMENT :

- In future we have a tendency to planned to construct our intelligent system during a compact size and additionally as globally acceptable to notify the No entry and No parking areas.
- if the rider enters in such area this system would immediately intimate as No entry area and vehicle can stop automatically. in case of any accident it might send the messages

to the friends continuously about the location of the accident happened until the first aid reaches the rider.

- Our system helps to know the location of the vehicle for rescuing in the case of theft incidents.
- This safety system technology can further be enhanced into four wheelers also by replacing helmet with seatbelt.

CONCLUSION:

The **conclusion** of this paper is to prevent the accident while driving in the vehicle. In this paper we are introducing a **smart helmet** which is used to **detect alcohol** consumption, usage of mobile phones while driving. In this we are insisting that every bike riders must wear the **helmet**.

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- [4] Yue –Cheng Wu, Yun-qing Xia &, Zhegiang, —Multichannel reflective PPG earpiece sensor with passive motion cancellation|| Biomedical Circuits &System, IEEE, 2007, PP 235-.....241.